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# The 2026 Embedded Hardware & Firmware Design Checklist

*Engineering Guidelines: From Schematic Review to Volume Production Ready*

## 1. Schematic Design & Component Selection

[ ] Verify pin multiplexing: ensure SPI, I2C, and UART peripherals do not clash with boot mode pins.

[ ] Decoupling Capacitors: Place 0.1uF ceramic caps as close as possible to every VDD pin with minimal via inductance.

[ ] Pull-up / Pull-down resistors: Verify open-drain I2C bus values (2.2k - 4.7k ohm depending on capacitive load).

[ ] Reverse polarity and ESD protection: Include TVS diodes on USB, RS-485, and user-accessible I/O headers.

[ ] Second-source availability: Ensure critical microcontrollers and regulators have multiple distributors.

## 2. PCB Layout & High-Speed Signal Integrity

[ ] Continuous reference ground plane directly beneath high-speed digital and RF traces.

[ ] Differential pair routing: Impedance match USB (90 ohm) and Ethernet (100 ohm) within 5 mil length delta.

[ ] Thermal relief: Adequate thermal vias beneath power switching ICs, LDOs, and MOSFET pads.

[ ] IPC-2221 trace width compliance for current rating and max allowable temperature rise (Delta T < 10°C).

[ ] Clear 30-mil clearance from PCB edge for all components to prevent panel depaneling stress damage.

## 3. Firmware Architecture, RTOS & Security

[ ] Hardware Watchdog Timer: Implement independent WDT refresh with timeout safe-mode recovery.

[ ] Secure Boot & Signature Verification: Flash digital signature verification before loading application code.

[ ] Non-blocking I/O: Ensure sensor communication operates via DMA or ISR with ring-buffers.

[ ] Fault logging: Persistent crash logs stored in internal Flash or EEPROM with stack trace.

## 4. Design for Manufacturing (DFM) & Testing

[ ] Dedicated SWD / JTAG test pads (1.27mm or 2.54mm) for automated bed-of-nails pogo pin fixtures.

[ ] Fiducial markers: At least 3 fiducials placed on diagonal corners of the board for SMT pick-and-place cameras.

[ ] Bill of Materials (BOM) contains verified manufacturer part numbers (MPN) and active lifecycle status.

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